

Solução do Problema CYCLE - Solver HiGHS

Análise Computacional

6 de outubro de 2025

1 Informações do Problema

Informações do Problema:

- Nome: CYCLE
- Número de Variáveis: 2857
- Número de Restrições: 1903
- Inviabilidade Primal: 0.000e+00
- Inviabilidade Dual: 6.990e-08
- Valor Primal: -5.226e+00
- Valor Dual: 0.000e+00
- Gap: -1.187e-16
- Número de Iterações: 33

2 Variáveis Primais e Custos Reduzidos ($x \geq 0$ e $z \geq 0$)

Tabela 1: Variáveis primais (x) e custos reduzidos (z)
com valores ≥ 0 do problema CYCLE

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1	3.025e+02	14	1.108e-01
3	6.500e+01	15	1.457e-01
4	4.560e+02	17	1.111e+00
5	2.410e+02	18	1.146e+00
6	1.100e+01	19	1.000e+00
7	3.100e+01	20	1.000e+00
8	3.025e+02	21	9.367e-01
9	1.100e+01	22	1.277e-01
10	2.531e+01	23	1.607e-01
423	3.460e+02	24	1.304e-02
472	1.000e+01	25	1.429e-02
474	3.460e+02	27	1.277e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
484	1.689e+01	28	1.607e-01
485	1.515e+00	29	1.304e-02
488	4.324e-01	30	1.429e-02
491	1.355e+00	52	1.108e-01
492	4.706e+00	53	1.457e-01
494	1.417e+00	110	5.718e-02
496	3.870e+01	124	1.457e-01
500	4.191e+00	125	1.108e-01
512	5.079e+00	158	1.945e-02
514	4.729e-01	159	2.277e-02
516	8.087e+00	163	7.240e-03
523	2.929e+00	165	1.111e-02
525	5.014e+00	166	1.422e-03
588	1.071e+01	167	1.199e-01
640	1.125e+00	176	1.945e-02
646	3.031e+00	177	2.277e-02
655	4.318e-01	181	7.240e-03
656	3.933e+00	183	1.111e-02
660	8.938e-01	184	1.422e-03
671	1.313e+00	185	1.199e-01
674	3.581e+01	193	1.006e+00
679	9.338e-01	194	1.009e+00
680	2.576e-01	195	1.000e+00
683	8.102e-01	196	1.008e+00
689	2.008e-02	197	9.940e-01
690	1.379e+00	198	1.011e+00
691	6.226e+00	199	1.011e+00
702	6.667e-01	200	1.127e+00
707	7.491e-01	201	1.004e+00
709	3.975e+01	202	9.857e-01
710	1.555e+00	203	9.055e-01
734	5.480e+01	204	8.744e-01
735	3.589e-01	205	1.003e+00
737	4.051e+02	206	1.009e+00
739	2.823e+01	207	1.001e+00
742	7.224e+01	208	1.004e+00
743	6.060e+00	209	1.004e+00
744	2.147e+00	210	1.003e+00
748	1.289e+00	211	8.548e-01
749	1.708e+00	212	9.203e-01
750	7.376e+00	213	2.256e-02
751	9.564e+00	214	2.385e-02
752	2.003e+01	215	1.304e-02
753	1.335e+02	216	6.160e-02
754	5.177e+01	217	8.250e-03
755	1.827e+01	218	2.495e-02
756	6.567e+00	219	5.039e-02
757	4.714e+00	220	1.564e-01
758	5.764e-01	221	1.549e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
759	1.297e+00	222	4.905e-02
760	4.131e-01	226	2.373e-02
762	1.197e+01	227	6.884e-02
764	1.515e+01	228	4.261e-02
765	1.197e+01	229	6.794e-02
766	3.572e+01	230	6.197e-02
767	5.086e+00	232	1.155e-02
768	6.320e+00	233	2.256e-02
769	1.776e+00	234	2.385e-02
770	1.783e+00	235	1.304e-02
771	1.340e+01	236	6.160e-02
772	1.122e+01	237	8.250e-03
775	3.708e-01	238	2.495e-02
776	3.648e-01	239	5.039e-02
778	3.643e+01	240	1.564e-01
779	4.081e+01	241	1.549e-02
794	2.975e+01	242	4.905e-02
795	3.300e+01	246	2.373e-02
796	2.975e+01	247	6.884e-02
797	3.383e+01	248	4.261e-02
798	8.660e+01	249	6.794e-02
799	1.129e+02	250	6.197e-02
800	8.660e+01	252	1.155e-02
801	3.025e+02	253	3.131e-02
802	1.100e+01	255	4.436e-02
803	2.531e+01	256	4.852e-02
804	4.477e+01	257	5.269e-02
806	4.519e-01	258	1.276e-01
821	3.025e+02	259	3.846e-02
822	1.100e+01	260	5.269e-02
823	2.531e+01	261	2.096e-02
825	2.803e+01	262	4.847e-03
827	8.973e+02	263	7.049e-03
829	6.512e+01	265	3.448e-02
832	5.571e+00	266	4.590e-02
837	2.383e+00	267	1.878e-02
841	3.784e+00	268	2.793e-02
842	3.834e+00	269	1.098e-02
843	6.015e+00	271	5.002e-02
844	4.288e-01	272	2.686e-02
845	2.298e-01	273	3.087e-02
848	8.103e-01	274	1.004e-01
852	1.125e+00	275	1.312e-02
853	5.696e+00	276	1.768e-03
854	8.117e+00	277	3.332e-02
857	1.690e+01	278	3.742e-02
858	1.229e+02	279	8.520e-02
859	1.944e+01	281	7.035e-02
860	3.367e+00	283	1.791e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
861	1.641e+01	284	4.973e-02
863	1.379e+01	286	1.792e-02
864	8.148e-01	287	6.833e-03
865	4.280e+00	294	2.933e-03
866	3.558e+00	296	2.682e-03
867	5.764e-01	297	1.353e-02
869	1.408e+01	300	4.001e-02
870	1.297e+00	301	8.553e-02
871	4.131e-01	303	1.029e-02
873	8.472e+01	304	7.771e-03
874	6.920e+01	305	3.690e-01
875	2.550e+01	306	6.650e-01
876	2.158e+01	307	4.816e-01
878	1.071e+01	308	5.673e-01
880	1.182e+01	309	6.499e-06
881	9.415e+00	311	2.876e-05
882	6.091e+01	313	2.625e-02
883	4.675e+01	315	1.381e-01
884	9.634e-02	316	6.539e-02
885	5.277e+01	318	3.614e-01
886	4.273e+01	319	3.341e-01
889	3.383e+01	358	2.070e-02
890	1.000e+01	359	2.093e-02
891	4.058e+00	371	3.077e-02
892	1.684e+02	372	3.681e-02
893	3.572e+01	378	7.339e-03
894	5.086e+00	379	1.403e-02
897	1.016e+03	380	1.012e-02
898	1.135e+03	386	3.154e-03
899	5.198e+03	389	9.401e-03
900	5.783e+03	393	6.224e-03
1164	6.240e+00	404	2.634e-03
1166	5.497e+00	405	2.452e-03
1175	5.497e+00	406	8.111e-04
1176	6.240e+00	407	3.287e-04
1216	2.159e+02	408	2.855e-03
1261	5.497e+00	419	5.032e-03
1267	2.159e+02	420	5.631e-03
1268	6.240e+00	421	1.587e-04
1276	6.500e+01	439	8.796e-03
1278	5.695e+00	440	5.657e-03
1280	5.428e+00	503	7.035e-03
1289	1.209e+00	507	6.027e-02
1292	1.001e+01	509	3.731e-02
1295	9.430e-01	530	7.035e-03
1297	9.355e-03	534	6.027e-02
1309	5.230e+00	536	3.731e-02
1310	3.476e+00	537	1.000e+00
1312	8.860e-01	538	1.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1317	3.609e+00	539	1.000e+00
1426	2.409e+00	540	1.000e+00
1452	1.190e+01	541	1.000e+00
1464	4.365e-03	542	1.000e+00
1477	6.131e-01	543	1.000e+00
1479	3.125e-02	544	1.000e+00
1481	2.955e-01	545	1.000e+00
1484	4.429e+00	546	1.000e+00
1485	3.498e-01	547	1.000e+00
1487	2.960e+00	548	1.000e+00
1490	6.822e-01	549	1.000e+00
1491	1.750e+00	550	1.000e+00
1495	3.972e-01	551	1.000e+00
1497	2.012e-01	552	1.000e+00
1502	1.498e+00	553	1.000e+00
1515	4.146e+00	554	1.011e+00
1516	2.654e+00	555	1.000e+00
1544	1.061e+00	556	1.003e+00
1548	4.987e-01	557	1.000e+00
1550	8.021e+01	558	1.000e+00
1551	1.279e+01	559	9.821e-01
1555	1.713e+01	560	1.000e+00
1556	1.132e+01	561	1.000e+00
1558	5.530e+00	562	1.000e+00
1559	4.394e+00	563	1.000e+00
1560	5.008e+00	564	1.060e+00
1561	5.417e-01	565	1.000e+00
1562	5.530e+00	566	1.037e+00
1563	4.017e+00	567	8.548e-01
1564	1.240e+01	568	9.088e-01
1565	3.868e+01	569	1.691e-02
1566	2.396e+01	570	1.591e-02
1568	5.010e-01	571	1.506e-02
1569	1.132e+01	572	1.306e-02
1571	5.135e+00	573	2.257e-02
1574	1.003e+01	574	1.304e-02
1575	1.406e+00	575	2.172e-02
1576	1.864e+00	576	1.654e-02
1577	5.473e-01	577	2.593e-02
1578	3.073e+00	578	5.398e-02
1579	2.435e+00	579	1.425e-02
1580	4.268e+00	580	1.429e-02
1581	1.692e-01	581	3.900e-02
1584	1.466e+01	582	2.938e-02
1585	1.003e+01	583	1.138e-02
1600	7.167e+00	584	6.335e-02
1601	6.500e+01	585	9.451e-02
1603	5.695e+00	586	1.569e-01
1605	4.016e-01	587	1.256e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1606	2.105e-01	589	1.433e-02
1623	6.500e+01	590	6.531e-02
1625	5.695e+00	592	6.810e-02
1626	1.306e+01	593	3.868e-02
1635	1.575e+01	594	6.416e-02
1637	8.068e+00	595	4.760e-02
1638	2.407e+00	596	8.484e-02
1643	1.329e+04	597	3.960e-02
1645	3.707e+00	598	9.633e-02
1646	3.614e+00	601	1.691e-02
1647	4.119e+00	602	1.591e-02
1648	1.178e+00	603	1.506e-02
1649	6.479e-01	604	1.306e-02
1650	1.336e+00	605	2.257e-02
1651	2.037e-01	606	1.304e-02
1652	1.071e-01	607	2.172e-02
1655	3.502e-01	608	1.654e-02
1658	3.708e+00	609	2.593e-02
1660	3.508e+00	610	5.398e-02
1662	5.262e+00	611	1.425e-02
1663	5.446e+00	612	1.429e-02
1664	3.688e+01	613	3.900e-02
1665	8.664e+00	614	2.938e-02
1666	1.455e+00	615	1.138e-02
1667	7.951e+00	616	6.335e-02
1673	8.007e-01	617	9.451e-02
1674	5.010e-01	618	1.569e-01
1675	1.574e+01	619	1.256e-01
1676	3.600e+00	620	1.388e-17
1677	1.132e+01	621	1.433e-02
1679	5.135e+00	622	6.531e-02
1681	2.146e+01	623	1.388e-17
1682	1.760e+01	624	6.810e-02
1683	1.554e+01	625	3.868e-02
1684	1.320e+01	626	6.416e-02
1687	2.409e+00	627	4.760e-02
1688	1.465e+01	628	8.484e-02
1689	1.190e+01	629	3.960e-02
1690	1.855e+01	630	9.633e-02
1691	1.321e+01	631	1.388e-17
1692	3.353e-02	632	1.648e-17
1693	8.381e+00	664	7.035e-03
1694	6.801e+00	696	7.035e-03
1697	7.167e+00	712	7.024e-03
1698	4.000e+02	807	5.056e-04
1701	1.329e+04	809	1.056e-03
1703	1.003e+01	810	5.169e-03
1706	1.287e+03	811	3.087e-03
1707	1.465e+03	812	4.909e-03

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1708	1.578e+03	817	2.760e-03
1709	1.706e+03	906	5.411e-02
1710	4.560e+02	907	5.449e-02
1714	4.271e+01	908	2.291e-01
1718	1.424e+01	909	9.888e-04
1723	2.110e+01	912	4.549e-03
1740	1.506e+01	913	2.335e-05
1747	1.892e+01	917	1.572e-04
1799	1.946e+01	918	1.506e-01
1827	2.320e+01	927	2.764e-02
1853	4.366e+01	930	5.411e-02
1875	2.711e+00	931	5.449e-02
1877	1.382e-01	932	2.291e-01
1882	8.468e+01	933	9.888e-04
1883	1.284e+01	936	4.549e-03
1886	2.812e+01	937	2.335e-05
1897	1.511e-01	941	1.572e-04
1904	5.797e-02	942	1.506e-01
1907	3.102e+00	951	2.764e-02
1910	1.027e+00	952	1.011e+00
1938	5.316e+02	953	1.012e+00
1943	1.752e+00	954	1.006e+00
1944	1.862e+02	955	1.006e+00
1947	2.445e+01	956	1.251e+00
1948	5.355e+01	957	1.004e+00
1951	2.928e+01	958	1.000e+00
1954	6.882e+01	959	1.028e+00
1955	2.324e+02	960	1.022e+00
1956	1.927e+02	961	9.875e-01
1960	4.008e+00	962	1.030e+00
1963	2.872e+00	963	1.031e+00
1964	2.517e+00	964	1.028e+00
1965	2.665e+01	965	1.171e+00
1966	1.029e+01	966	1.012e+00
1967	1.285e+01	967	9.859e-01
1968	2.459e+00	968	8.913e-01
1969	2.630e+00	969	8.761e-01
1970	6.470e+00	970	1.004e+00
1971	1.114e+01	971	1.020e+00
1976	3.900e+01	972	1.001e+00
1977	3.694e+01	973	1.004e+00
1981	1.891e+01	974	1.003e+00
1984	2.424e+01	975	1.011e+00
1985	1.891e+01	976	1.011e+00
1993	6.852e+01	977	1.014e+00
1994	6.683e+01	978	1.016e+00
1995	6.852e+01	979	1.036e+00
1996	4.560e+02	980	8.576e-01
2014	4.560e+02	981	9.105e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
2020	1.608e+00	982	2.507e-02
2021	1.935e+01	983	2.126e-02
2022	1.503e+02	984	2.536e-02
2027	1.614e+01	985	1.544e-02
2028	4.325e+01	986	2.638e-01
2030	1.112e+01	988	2.223e-02
2031	7.637e+00	989	4.879e-02
2032	1.169e+01	990	7.390e-02
2037	1.934e+01	992	3.967e-02
2040	1.531e+01	993	3.858e-02
2041	4.305e+01	994	6.666e-02
2042	2.204e+02	995	1.986e-01
2043	7.300e+01	996	2.136e-02
2044	7.029e+01	997	5.942e-02
2051	4.008e+00	998	4.445e-03
2053	3.638e-01	1001	4.164e-02
2056	2.872e+00	1002	6.608e-03
2057	2.517e+00	1003	6.900e-02
2058	9.504e+01	1004	7.280e-02
2059	7.805e+01	1005	4.586e-02
2060	4.114e+01	1006	7.229e-02
2061	3.398e+01	1007	7.165e-02
2063	1.946e+01	1008	7.177e-02
2064	2.320e+01	1009	9.330e-02
2065	5.434e+01	1012	2.596e-02
2066	4.366e+01	1013	2.295e-02
2067	1.729e+02	1014	2.484e-02
2068	1.286e+02	1015	1.699e-02
2069	7.518e-03	1016	2.654e-01
2070	5.203e+00	1017	1.775e-03
2071	4.187e+00	1018	2.372e-02
2072	2.303e+00	1019	5.018e-02
2075	1.000e+01	1020	7.567e-02
2077	5.689e+00	1021	1.712e-03
2078	2.361e+02	1022	4.119e-02
2079	2.665e+01	1023	4.033e-02
2080	1.029e+01	1024	6.705e-02
2083	4.780e+03	1025	2.003e-01
2084	5.434e+03	1026	2.308e-02
2085	1.487e+04	1027	5.744e-02
2086	1.643e+04	1029	1.612e-03
2087	2.410e+02	1030	1.715e-03
2266	1.739e+01	1031	3.976e-02
2268	8.230e+00	1032	7.989e-03
2269	1.455e+01	1033	7.066e-02
2271	1.931e+00	1034	7.393e-02
2276	9.677e-01	1035	4.735e-02
2295	7.928e+00	1036	7.359e-02
2305	3.725e+00	1037	7.334e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
2468	6.509e+00	1038	7.313e-02
2470	3.460e+01	1039	9.507e-02
2471	1.478e+00	1040	9.996e-04
2477	3.813e+00	1041	1.760e-03
2488	6.420e+00	1059	3.612e-02
2500	2.300e+01	1062	3.438e-02
2502	5.785e+00	1067	3.175e-03
2503	3.925e+01	1070	3.232e-02
2532	2.699e+02	1088	3.612e-02
2533	6.476e+01	1091	3.438e-02
2537	7.372e+01	1096	3.175e-03
2538	2.447e+01	1099	3.232e-02
2542	2.966e+01	1100	2.891e-01
2544	2.234e+00	1101	5.326e-01
2545	9.944e+00	1102	3.688e-01
2547	1.656e+01	1103	3.730e-01
2548	3.947e+01	1104	4.381e-01
2549	1.525e+02	1106	3.291e-05
2550	5.360e+01	1107	1.988e-03
2554	3.715e+00	1108	4.059e-02
2555	7.449e-01	1110	1.751e-01
2557	1.998e+00	1111	6.199e-02
2559	3.094e+01	1113	3.028e-01
2560	4.946e+00	1114	2.715e-01
2561	3.718e+00	1173	9.113e-04
2562	3.596e+00	1174	3.175e-03
2563	2.540e+00	1180	2.534e-02
2564	3.348e+00	1181	6.556e-03
2565	1.966e+01	1182	1.305e-02
2568	7.906e-01	1188	9.341e-05
2569	9.157e-01	1198	6.927e-05
2570	5.282e-01	1201	3.768e-04
2571	3.634e+01	1202	1.256e-03
2572	3.589e+01	1204	1.092e-03
2588	1.545e+01	1293	8.811e-03
2589	2.410e+02	1294	1.923e-01
2611	2.410e+02	1299	6.972e-04
2614	6.798e+01	1300	1.431e-02
2623	6.859e+01	1301	5.226e-02
2625	1.300e+01	1302	1.321e+00
2626	1.020e+01	1320	8.811e-03
2631	7.545e+02	1321	1.923e-01
2634	2.404e+01	1326	6.972e-04
2636	3.893e+00	1327	1.431e-02
2637	3.928e+00	1328	5.226e-02
2638	4.804e+00	1329	1.321e+00
2639	1.074e+00	1333	1.000e+00
2640	5.574e-01	1334	1.000e+00
2643	1.484e+00	1335	1.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
2646	6.509e+00	1336	1.000e+00
2648	1.486e+01	1337	1.000e+00
2650	1.639e+01	1338	1.000e+00
2651	1.756e+01	1339	1.000e+00
2652	1.496e+02	1340	1.000e+00
2653	2.457e+01	1341	1.000e+00
2654	6.166e+00	1342	1.000e+00
2655	9.483e+00	1343	1.000e+00
2662	3.715e+00	1344	1.000e+00
2665	7.449e-01	1345	9.629e-01
2667	1.998e+00	1346	1.192e+00
2669	5.046e+01	1347	1.000e+00
2670	4.307e+01	1348	1.000e+00
2671	1.382e+01	1349	1.000e+00
2672	1.165e+01	1350	8.306e-01
2678	7.083e+01	1351	8.744e-01
2679	5.282e+01	1352	1.003e+00
2681	8.439e+01	1353	1.000e+00
2682	6.804e+01	1354	1.294e+00
2684	1.510e+01	1355	1.000e+00
2686	1.000e+01	1356	1.000e+00
2688	1.437e+00	1357	1.000e+00
2689	8.141e+02	1358	1.325e+00
2690	3.094e+01	1359	1.000e+00
2691	4.946e+00	1360	1.000e+00
2696	5.921e+03	1361	1.001e+00
2697	6.516e+03	1362	1.014e+00
2698	7.800e+00	1363	1.052e+00
2699	1.510e+01	1364	2.321e+00
2700	4.100e+01	1365	1.000e+00
2716	1.500e+01	1366	1.000e+00
2717	2.517e+02	1367	1.000e+00
2718	9.600e+01	1368	8.614e-01
2719	1.300e+01	1369	1.035e+00
2720	3.300e+01	1370	2.320e-02
2721	2.800e+01	1371	1.923e-02
2723	8.000e+01	1372	2.508e-02
2724	3.200e+02	1373	1.766e-02
2725	1.370e+02	1374	2.555e-02
2726	2.400e+01	1375	1.462e-02
2728	3.300e+01	1376	2.656e-02
2729	3.067e+01	1377	2.604e-02
2730	7.380e+00	1378	5.398e-02
2732	1.129e+02	1379	1.425e-02
2733	8.783e+00	1380	1.823e-02
2734	6.683e+01	1381	1.692e-02
2735	1.300e+01	1383	2.217e-01
2766	1.500e+01	1384	1.138e-02
2767	8.472e+01	1385	8.342e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
2768	2.550e+01	1386	1.121e-01
2770	1.173e+01	1387	1.388e-17
2772	1.170e+01	1390	6.760e-02
2773	6.031e+01	1391	3.185e-01
2774	5.251e+01	1392	8.315e-02
2777	3.300e+01	1393	5.585e-02
2778	7.380e+00	1394	6.548e-02
2780	1.129e+02	1395	3.312e-01
2781	5.752e-01	1396	6.896e-02
2799	2.146e+01	1397	7.799e-02
2800	1.554e+01	1398	4.172e-02
2803	2.640e+00	1399	7.547e-02
2804	1.450e+01	1400	8.772e-02
2805	1.836e+01	1401	1.361e+00
2806	8.339e+00	1402	5.778e-02
2810	3.938e+00	1403	5.415e-02
2813	5.000e-01	1404	5.902e-02
2816	9.504e+01	1406	1.260e-01
2817	4.114e+01	1407	2.320e-02
2819	2.127e+01	1408	1.923e-02
2820	2.536e+01	1409	2.508e-02
2821	5.380e+01	1410	1.766e-02
2822	1.712e+02	1411	2.555e-02
2823	5.177e+00	1412	1.462e-02
2824	2.400e+01	1413	2.656e-02
2828	4.000e+00	1414	2.604e-02
2829	6.683e+01	1415	5.398e-02
2844	5.046e+01	1416	1.425e-02
2845	1.382e+01	1417	1.823e-02
2849	7.012e+01	1418	1.692e-02
2850	8.397e+01	1420	2.217e-01
2854	2.673e+01	1421	1.138e-02
2856	3.707e+00	1422	8.342e-02
		1423	1.121e-01
		1427	6.760e-02
		1428	3.185e-01
		1429	8.315e-02
		1430	5.585e-02
		1431	6.548e-02
		1432	3.312e-01
		1433	6.896e-02
		1434	7.799e-02
		1435	4.172e-02
		1436	7.547e-02
		1437	8.772e-02
		1438	1.361e+00
		1439	5.778e-02

3 Variáveis Duais (Multiplicadores de Lagrange - Valores λ 0)

Tabela 2: Variáveis duais (y) com valores λ 0 do problema CYCLE

Coordenada y	Valor y (Dual)
7	1.115e-03
10	1.115e-03
12	1.108e-01
14	1.457e-01
66	1.280e-01
108	1.514e-03
112	8.840e-03
175	1.115e-03
176	5.657e-03
178	8.796e-03
186	7.622e-03
214	1.066e-02
215	1.066e-02
216	1.060e-02
217	1.225e-02
219	1.095e-02
220	1.057e-02
221	1.138e-02
224	1.270e-01
225	4.109e-03
229	1.797e-02
232	1.625e-02
233	2.070e-02
234	2.093e-02
236	1.784e-02
237	1.907e-02
240	1.974e-02
241	1.974e-02
244	1.326e-02
245	1.269e-02
246	1.798e-02
247	2.852e-02
248	1.208e-02
249	1.603e-02
252	1.974e-02
253	2.621e-02
264	2.635e-03
267	3.077e-02
268	3.681e-02
269	9.401e-03
271	7.406e-04
273	2.770e-03
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Coordenada y	Valor y (Dual)
274	1.895e-02
275	1.135e-02
276	2.009e-01
277	3.931e-03
278	3.779e-03
281	2.954e-03
286	1.023e-02
287	1.023e-02
290	4.332e+00
373	2.258e-03
378	9.860e-04
396	2.258e-03
398	9.860e-04
419	1.115e-03
505	1.331e-01
513	1.080e-02
515	2.635e-03
529	7.035e-03
530	7.024e-03
535	6.027e-02
537	3.731e-02
583	2.356e-03
602	5.308e-03
652	5.308e-03
679	1.115e-03
682	1.115e-03
683	1.149e-02
684	1.152e-02
685	6.425e-03
686	6.469e-03
687	2.509e-01
688	4.354e-03
691	2.768e-02
692	2.170e-02
719	3.049e-02
720	3.052e-02
721	3.030e-02
722	1.498e-02
724	3.126e-02
725	3.091e-02
726	2.823e-02
729	1.710e-01
730	1.170e-02
748	3.612e-02
751	3.438e-02
769	4.350e-03
772	9.113e-04
773	3.175e-03
774	1.980e-02
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Coordenada y	Valor y (Dual)
775	1.246e-03
776	3.671e-03
777	3.451e-03
781	3.232e-02
782	1.505e-01
783	1.123e-02
784	1.075e-02
785	1.405e-02
786	1.571e-02
787	3.605e-02
790	2.913e-02
791	2.913e-02
794	9.475e-01
921	1.115e-03
924	1.115e-03
971	8.811e-03
975	1.923e-01
1025	2.635e-03
1031	2.944e-01
1035	3.247e-01
1045	6.972e-04
1046	1.431e-02
1047	5.226e-02
1048	1.321e+00
1053	3.476e-02
1186	1.115e-03
1189	1.115e-03
1238	1.923e-01
1259	1.154e+00
1270	1.260e-01
1280	2.635e-03
1283	2.912e-02
1300	3.831e-02
1343	2.229e-03
1363	5.023e-03
1411	5.023e-03
1438	1.115e-03
1440	1.034e-01
1441	1.044e-01
1442	6.746e-02
1443	6.844e-02
1467	1.495e-01
1468	1.500e-01
1469	1.489e-01
1470	3.048e-01
1484	1.776e-02
1498	3.422e+01
1501	1.744e-01
1502	1.744e-01

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Coordenada y	Valor y (Dual)
1504	2.272e-01
1506	4.925e-03
1608	1.115e-03
1619	2.943e-02
1620	3.006e-02
1658	1.923e-01
1706	2.635e-03
1712	3.667e-01
1716	4.107e-01
1726	9.910e-04
1727	1.679e-02
1728	6.427e-02
1729	1.708e+00
1730	1.288e-02
1732	4.559e-02